

Module title

Determinants of Health and Disease and Research Methodology

Course #: COMH2022

Has three module components

- Environmental Health
 - Behavioral Science
 - Research Methods
- **Period: March 20 – April 14, 2023 – 4 weeks**
 - Module requirement: Pre-med courses
 - Placement: Year II, PC 2, Semester I
 - **ECTS 5 (6?),**

Learning Outcomes

- ***At the end of this module, medical students will be able to:***
 - Analyze **determinants of health** and disease broadly at individual, family, and community levels and mobilize them for **action**;
 - Collect or develop and use **effective communication skills** with appropriate communication technology to transfer health-related information to assist in health **promotion and disease prevention actions**;
 - Critically evaluate medical literature and develop a **research proposal, and be able to implement**;

Graduate profile (Core competencies)

1. Medical expert 2. Communicator 3. Collaborator 4. Manager	5. Health Advocate 6. Scholar and Researcher 7. Professional
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Medical Experts

- Understand the normal and abnormalities in body structure/ function
- Recognize the normal and abnormal human behavior.
- Understand the important determinants of health.
- Know the molecular, cellular, biochemical and physiological mechanisms that maintain the body's homeostasis.
- Understand the principles of drug action and its use, and efficacy of various therapies.

Health Advocate

- Advance the well-being of individual patients, communities, and populations.
- Identify advocacy activities as important activities for the individual patient, for populations of patients and for communities.
- Identify and collaboratively address broad health issues and the determinants of health.

Expectations from Instructors

- Provide guiding Lectures
- Facilitate discussions
- Encourage students participation
- Insist and arrange individual reading assignments
- Arrange and be available for consultancy/ advising

Expectation from Students;

- Avail in classes
- Ask questions as we go through the lectures
- Participate in classroom discussion/
exercises/
- Participate in group works
- Complete reading assignments
- Actively participate in term paper works

Schedule for PUBH 2022

Module IX, ECTS 6

Weeks	Days Assigned	Courses and Activities Assigned
Week 1	March 20-29	Environmental Health (EH)
Week 2	March 30-05	Behavioural Health Science (BHS)
Week 3	April 06-12	Research Methodology (RM)
Week 4	April 13	Coursework/assignment presentations and Reading day
	April 14	Final Examination day

Course Coordinator

- The COMH2022 -Kalkidan Solomon:
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 - EH Coordinator: Meaza Gezu
 - Email meaza.gezu@gmail.com
- Office: SeferesSalem Campus, 6th floor

EH Instructors

1. Abera Kumie, MSc, PhD, aberkumie2@yahoo.com
2. Samson Wakuma, PhD: samson_wakuma@yahoo.com
3. Teferi Abegaz, PhD, teferiabegaz@gmail.com
4. Meaza Gezu, Candidate PhD, meaza.gezu@gmail.com
5. HaileMichae Mulugeta: hailumary464@gmail

EH Sub-Module

Course Description: Contents

- Objectives
- Scope
- Methods
- Assessment
- Timing
- Assignment
- References

EH course Learning Outcomes

- Identify EH determinants of health and disease broadly at individual, family, and community levels;
- Analyze environmental and ecological determinants of health and disease at individual, family, and community levels;
- Help individuals and families to identify actions to prevent and control environmental and ecological determinants of ill health;

Course structure

- General related issues / background
- The status from Global to local approach
- Description of “Healthy” and “unhealthy” Environment
- Causes/ factors of adverse effects
- Evaluation principles (What is normal and not: pollution, safety...)
- Prevention and Control Strategies

Scope of Environmental Health Sessions

- Biosphere and pollution
- Human – environment interaction
- Environmental Health and Ecology of health and disease
- Describing What Determinants of Health is

UG_EH Course Schedule: March 20-29, 2023

7 Days

Date	Sessions	Time	Instructor name
March 20	Course description and introduction	1hr (9am-9:50am)	Dr Abera K.
March 20 (M)	Introduction to environmental health; Principles of pollution and its control	• 3hrs (10am-11:50am) • 2pm-2:50-pm	Dr Abera K.
March 21 (Tue)	Ambient and indoor air pollution and water pollution its control	3hrs (9am-11:50am)	Dr. Samson W.
March 22 (Wed)	Introduction to ecology and environmental health, Irrigation, water development and PH relevance sociological environment (Health and culture)	1.5 hrs (9am-10:30am)	Dr. Teferi A.
March 22	Municipal waste management and medical waste management and infection prevention	1.5 hrs (10:30am-12:00am)	Dr. Teferi A.

UG_EH Course Schedule: March 20-29, 2023

March 23 ((Thur)	Provision of safe and adequate water supply	2 hrs (9am-10:50am)	Dr Abera K
March 23 (Fri)	Human waste management, technical country status and management	2 hrs. (11:00-11:50am) (2pm-2:50-pm)	Dr Abera K.
March 24 (Fri)	Global and environmental health problems site and location: altitude, climate, and health	2hrs((9am-10:50am)	Dr. Samson W.
March 27	Occupational health and safety	2hrs (9 am-10:50am)	Dr. Teferi A.
March 28	Housing and health	1hrs (9am-09:50am)	Hailemichael
March 28	Public health important vectors and their control	1hr ((10:00am-11:50pm)	Hailemichael
March 29	Food hygiene	2 hrs (10pm-11:50pm)	Hailemichael
April 13	EH term paper presentation	3hrs (9am-11:50am)	All

Module requirements

- Courses
 - Infectious Diseases: INFD2121
 - Measurement of Health & Diseases (COMH1012) Yr I, Sem II)
 - Epi Course?
- Registration of the Modules
- Attendance- Complete
- Lecture Note-taking: in addition to having copies of slide presentations

Methods of instruction

1. Lecture presentation
2. Discussions/questioning/comments
3. Individual assignments exercises
4. Group work exercises

Course Grading

- Progressive assessment/ **Formative**
 - Attendance, participation: 10%
 - Group work: grading 30%
- **Summative**
 - Written exam: 60% (twice Part I and Part II, each 30%)
 - Short answers, multiple choices, true/ false

Student's assessment

- Participating to improve the course
 - Feedbacks are encouraged
 - Any time in the course of teaching
 - At the end of the course

On the content of

- Course structure
- Course materials
- Mode of learning-teaching

Class Policy

- Attendance is compulsory
- Late coming is very much discouraged
- Participation is highly encouraged
- Cheating/ plagiarism is prohibited

Course materials

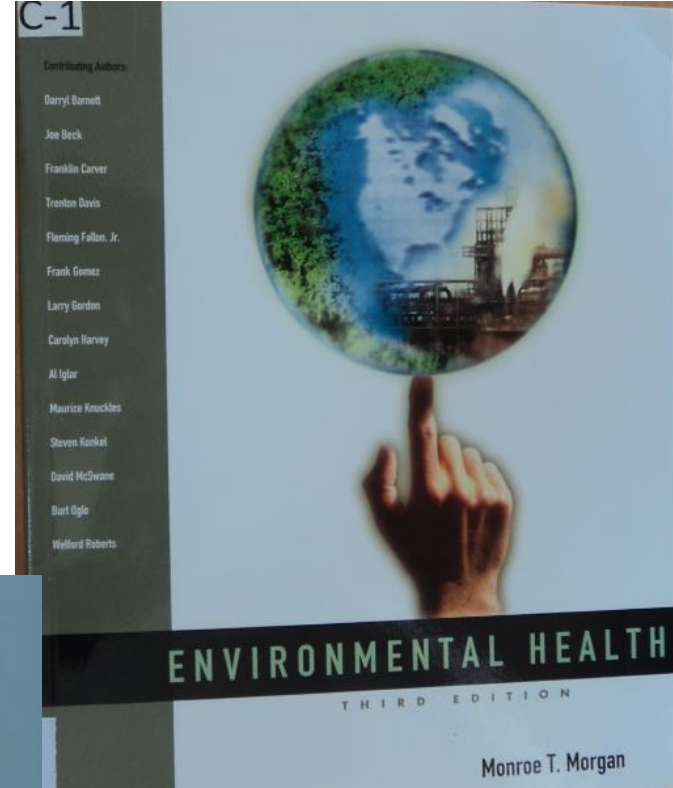
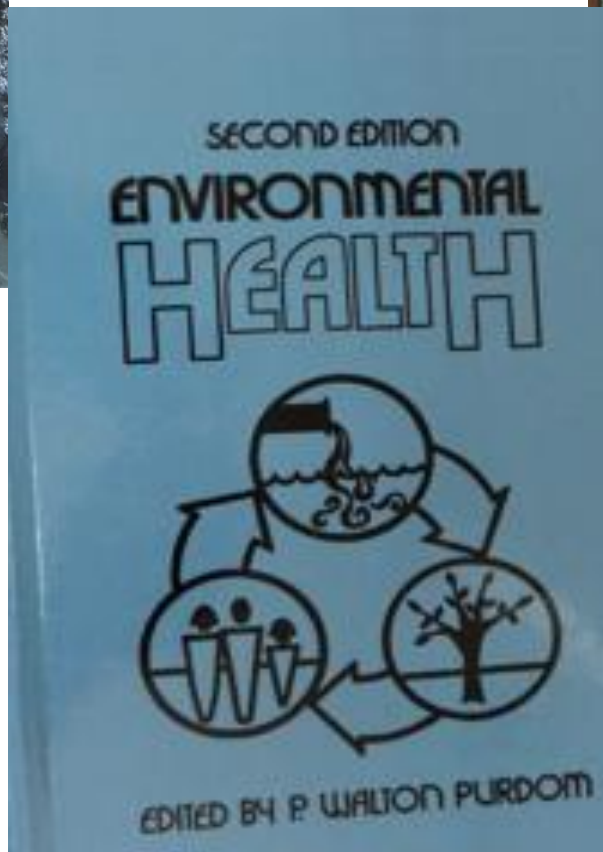
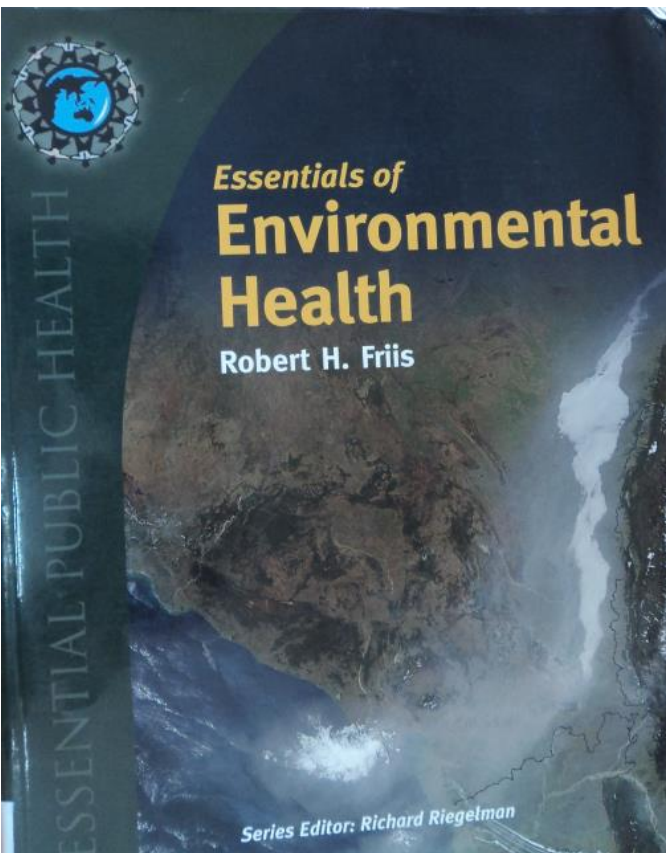
- Lecture notes: pdf slides-Soft copies will be available from instructors
- Reading text books
 - (available in the library)
- Soft copies: will be available in folders
- Browsing internet: students can use

Reading materials: Text books - Library

- Purdom PW, editor. Environmental health. San Diego: Academic press, 2nd Edition, 1980
- Essentials of Environmental Health Robert Friis, 2007
- Environmental Health Edited by Monroe T Morgan, 2003

Reading text books: additional (Soft copies will be available)

- Salvato JA. Environmental Engineering and Sanitation, 3rd edition, New York, John Wiley & Sons, 1982; 1990
- William P. Cunningham, Mary Ann Cunningham. Principles of environmental science: inquiry and applications, sixth edition, 2011



END OF PRESENTATION

Any Q?

Any comment
on this picture
Vs EH factors

Assignments

- Group Assignments: Term paper
- This will be presented

EH Term paper assignments 30%

Deliverables

- Presentation and submission of slides
- Date is April 13 /2023
- Place of presentation: Classroom-based for each gp

Title selection

- **Term paper topics selection**
- By own?
- Be Offered

Consider the issue of the title:

- Not too broad
- Feasibility
- Relevance
- Acceptability
- Available resource

3 topics per group

- Select 1 after consultation
- Further develop the idea with your advisor

Project Paper Evaluation

- A total of 30 minutes
 - 20 presentations;
 - 10 discussions;
- ***Presentation evaluation will be based on the areas:***
 - Organization;
 - Ideas flow;
 - Clarity;
 - Objectives coverage;
 - Ability of discuss and conclusions;
 - Confidence;
 - Performance in defending challenges;
 - Use of references;
 - Timing is well evaluated;

Grouping

SN	Term Paper Title/ Advisor	Gp lead and Mobile	Gp #ing
1	Is Addis Ababa clean-Challenges and Opportunities: observation in one of Kebeles around “Autubus Tera” (Advisor-Abera K)		
2	Is Addis Ababa clean-Challenges and Opportunities: observation in one of Kebeles around SefereSelam (Advisor-Abera K)		
3	Plastic Waste Recycling by Addis Ababa Municipality (Advisor: Samson W)		
4	Waste Recycling practice in Minalech Tera, Markato by Addis Ababa Municipality (Advisor: Samson W)		
5	Public latrine Management in Addis Ababa: use and challenges in Addis Ketema (Advisor: HaileMichael)		
6	Health care waste management in TASH (Advisor Teferi A)		

How to proceed for term paper work;

It requires a structured plan on less than 1 page:

- Background (why the topic was selected)
- Objectives
- Methods and tools:
 - Source of data
 - Instruments (search criteria, search guides/ literature review guides
 - Activities by time (study proposal, data collection, data interpretation and write-up, presentation
- Identify/assign to the advisor for each title of the term paper
- Submission of a plan for advisors:

Term paper Guides

1. Is Addis Ababa clean-Challenges and Opportunities: observation in one of Kebeles around “Autubus Tera”
2. and around SefereSelam?

- Identify and describe the locations by mapping
- Identify the subjects: liquid and solid waste management
- Walk through the locations
- Document your findings in a text and Photos
- Ask someone why that happened, this is if needed
- Organize your data
- Prepare slides for the presentation
- Organize references used

Term paper Guides

3. Plastic Waste Recycling by Addis Ababa Municipality
4. Waste Recycling practice in Minalech Tera, Markato by Addis Ababa Municipality

- Visit a nearby sub-city, waste management Office
- Identify what types of plastics are recyclable
- Identify the institutions/ community involved in recycling
- Interview if needed
- Google the benefits of recycling
- Organize data
- Document your findings in slides and Photos
- Indicate any Suggestions
- Organize references used
- Prepare slides for the presentation

Term paper Guides

5. Municipal public latrine Management in Addis Ababa: use and challenges

- Identify at least 2 public latrines around “Autobus Tera”
- Describe the public latrine: how many seats, who are using it, how it is managed (who manages it? Any payment for use?); Is it clean? Is there water? ??
- Describe the challenges and opportunities
- Ask the opinion of users on the public latrine
- Document your findings in Photos and slides
- Any Suggestions
- Organize references used
- Prepare slides for the presentation

Term paper guides

6. Healthcare waste management in TASH laboratory

- Describe the type of waste?
- How stored: in the primary site? Final storage?
- Describing the color of the waste container
- How collected?
- How finally disposed of?
- Document your findings in Photos and slides
- Any Suggestions
- Organize references used
- Prepare slides for the presentation

What must the Gp do?

Six groups

- Meet together
- Select Chairperson and Reporter/Secretary
- Select the title
- Arrange division of labor: data collection, write-up, photo taking
- Discussing the findings
- Writing the presentations on slides
- Agreeing on who will present (not more than 2 students)
- Contact advisors/ mentors: before data collection and before presentation

Methods for the Term Paper

- Literature review
- Observation with a check list; Photo taking
- Interviewing

Maintaining quality of slides

- Not crowded
- Readable font size
- Slide numbering
- 1-2 photos/ images per slide
- Photos and images with caption, date, and place

Presentation Structure

(Max 20 minutes)

Outline

- Title
- Outline
- Introduction (background rationale)
- Objectives
- **Methods (explicitly)**
- Results and Discussion
- **Strengths** (What was good)
- **Limitations** (Challenges-What was bad in data collection)
- Conclusions
- Recommendations
- Acknowledgements
- **References: with Vancouver style (example)**

Powerpoint presentation format:

- Slide paging
- Not crowded
- Bulleted (no sentence)
- Font size: > 20
- Bullets: < 10 per slides
- Preferably 7 bullets/ slide
- Pictures: 1-2 per slide; better /slide
- Pictures with foot notes (title, place, date)
- **Total slides: 20-25.**
- **Slide writing: no shading, color use: use black color on white background**

END